MedIntel: A Multi-Agent AI Framework for Clinical Decision Support Using Machine Learning, Rule-Based Validation, and Knowledge Retrieval

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Abstract— Clinical decision support systems increasingly combine machine learning, knowledge retrieval, and language-model reasoning, yet the integration of these components can become difficult to audit when a single generative model is responsible for the final recommendation. This paper presents MedIntel, a proposed multi-agent framework that treats patient analysis as a coordinated pipeline rather than a single prediction step. Its objective is an implementable, modular architecture for patient-level decision support in which every recommendation can be traced to the evidence that produced it. The framework accepts structured patient records and associated event or log data, validates and normalizes the input, applies task-specific machine-learning analysis, executes deterministic rule checks, retrieves supporting medical knowledge, and then combines the resulting evidence through a decision-fusion layer whose explicit confidence function weighs model, rule, and retrieval support against disagreement among components. A dedicated explanation and audit stage converts technical outputs into a structured report that can be reviewed by a clinician. The design also supports parallel processing of multiple patient records in isolated contexts and controlled re-indexing of validated knowledge. Unlike systems that rely mainly on free-form large language model (LLM) inference, MedIntel assigns explicit responsibilities to analytical components and reserves generative models for contextual synthesis and explanation. The present work is a design-stage study; therefore, no empirical accuracy or clinical-performance claims are made. Instead, an evaluation protocol is defined for future validation using benchmark datasets, progressively richer baselines, component ablation studies, calibration measures, and human review of explanation quality. By separating learned, rule-based, and retrieved evidence, the framework aims to make patient-level clinical reasoning more traceable and easier to audit.

Keywords— clinical decision support, multi-agent systems, retrieval-augmented generation, rule-based reasoning, explainable AI, healthcare AI

I. INTRODUCTION

MedIntel is a medical log analysis system for generating useful diagnostic results. Clinical decision support systems (CDSSs) have traditionally been built around rules, statistical models, or retrieval services that answer narrowly defined clinical questions [1]. MedAgents uses several role-based large language model (LLM) agents that independently analyze a question, discuss the findings, and reach a consensus [2]. ClinicalRAG similarly uses a multi-agent pipeline to connect clinical inputs with heterogeneous structured and unstructured knowledge [3]. KG4Diagnosis extends the idea by combining hierarchical agents with a medical knowledge graph for specialized diagnosis [4]. These systems can be useful because their inputs and outputs are comparatively structured, yet real hospital data rarely arrives in a perfectly normalized form. An ensemble combining complementary retrieval and predictive approaches can improve diagnostic ranking, including for less common conditions [5], [6]. Patient records may contain missing fields, inconsistent units, repeated observations, free-text notes, and event logs that describe the same clinical episode from different perspectives. The challenge is therefore not only to predict a condition, but also to decide which pieces of evidence should influence that prediction and how the resulting recommendation can be explained to a human reviewer.

The rapid development of LLMs has expanded the design space. Modern reviews of clinical LLM systems identify calibration, validation, and transparency as recurring weaknesses [7]. These weaknesses can also be seen broadly in retrieval-augmented generation (RAG) studies [8], [9]. Research on self-learned clinical knowledge has made the governance problem more visible [10]. A review by Thirunavukarasu et al. describes broad applications of LLMs in medicine while emphasizing the domain-specific risks and validation requirements that accompany clinical use [11]. Subsequent studies have shown that language models can support clinical decision tasks [12], especially when external knowledge is retrieved at inference time. Wang et al. discussed RAG as a mechanism for making model responses more specific to current clinical evidence [8], while Kresevic et al. demonstrated that structured guideline retrieval and careful prompt construction can improve the accuracy of guideline interpretation [9].

More recent work has moved from single-model prompting toward multi-agent systems. In 2025, Öğdü et al. proposed an adaptive architecture that couples specialist agents, biomedical RAG, web evidence, data fusion, and a confidence-triggered feedback loop [1]. These works demonstrate the value of decomposition and collaboration, but they also illustrate an important engineering trade-off: the more reasoning components are introduced, the more difficult it becomes to identify which component produced a particular recommendation.

MedIntel is motivated by a simpler principle: different kinds of evidence should be handled by the type of component that is best suited to that evidence. Machine-learning models are appropriate for learned patterns in structured features; rule engines are appropriate for explicit thresholds and institutional policies; retrieval systems are appropriate for contextual evidence; and language models are useful for synthesis and explanation. Rather than asking one model to perform every task, the proposed framework coordinates these components and preserves their outputs as an auditable evidence set.

The principal objective of this work is therefore to define an implementable, modular architecture for patient-level decision support that can process multiple patient records in parallel, combine learned and deterministic evidence, produce structured explanations, and preserve a trace of the inputs that led to the recommendation. Because the work is currently at the proposal stage, this paper is intentionally positioned as a design and methodology study. No experimental results are reported; instead, the paper specifies how the proposed system should be evaluated once implementation is complete.

II. LITERATURE REVIEW

Recent research can be organized around four complementary directions: LLM-based medical reasoning, retrieval-augmented clinical systems, multi-agent diagnostic frameworks, and predictive or rule-based decision support. The broader medical-LLM literature recognizes the promise of language models for clinical tasks while emphasizing risks related to factuality, context dependence, and clinical validation [11]. These limitations motivate the use of explicit evidence sources rather than unrestricted model memory.

A. LLM-Based Medical Reasoning MedAgents is a representative multi-agent approach in which role-specific language-model agents propose independent analyses, discuss the findings, summarize the discussion, and reach a decision [2]. The study shows why decomposition can help medical reasoning: different agents can focus on distinct perspectives before a final synthesis. MedIntel adopts the same general principle of specialization but moves the analytical responsibilities beyond LLMs by introducing dedicated machine-learning and deterministic-rule components.

B. Retrieval-Grounded Clinical Systems RAG-based approaches attempt to connect language-model outputs to external evidence. Wang et al. argued that retrieval can make clinical responses more specific to current medical information [8]. Kresevic et al. subsequently showed how structured guideline retrieval and careful prompt construction can substantially improve guideline interpretation for a focused clinical task [9]. Liu et al. conducted a systematic review and meta-analysis of biomedical RAG studies and reported an overall performance benefit over baseline LLMs, while also identifying methodological weaknesses that warrant stronger validation [6]. ClinicalRAG extended this direction through heterogeneous structured and unstructured medical knowledge in a multi-agent pipeline [3].

C. Knowledge-Structured and Hierarchical Agents KG4Diagnosis combines a general practitioner-style agent with specialized agents and an automatically constructed medical knowledge graph [4]. This provides a useful model for representing relationships between symptoms and diseases, but it also introduces additional knowledge-engineering requirements. The 2025 adaptive multi-agent CDSS by Öğdü et al. integrates biomedical RAG, web evidence, data fusion, and an adaptive feedback loop [1]. Their architecture is closely related to MedIntel in its emphasis on modularity and confidence-aware processing, but MedIntel is deliberately narrower: it places the learned prediction and rule engine at the center of patient-level structured-data analysis and uses retrieval and language generation as supporting layers.

D. Retrieval-Based Diagnosis and the Remaining Gap Abdullahi et al. found that an ensemble combining complementary retrieval and predictive approaches can improve diagnostic ranking, including for less common conditions [5]. Taken together, these studies suggest that a robust clinical-support pipeline benefits from multiple evidence sources, yet they do not provide a simple, auditable architecture in which tabular machine learning, deterministic rules, retrieval, and generative synthesis share a common patient-level interface. MedIntel addresses this engineering gap by defining clear contracts between these components rather than treating multi-agent behavior as unrestricted conversation. The resulting design is intentionally modest in algorithmic novelty and stronger in system integration, traceability, and reproducibility.

III. METHODOLOGY

This section formalizes the problem, describes the proposed MedIntel architecture, and defines the protocol for its implementation and evaluation.

A. Problem Formulation and Design Objectives Let a patient case be represented as a heterogeneous input $X = \{x_s, x_t, x_l\}$, where x_s denotes structured attributes such as age, laboratory values, vital signs, or categorical features; x_t denotes text-based clinical context; and x_l denotes related event or log observations. The system produces a candidate decision y, an evidence set E, and an explanation z. MedIntel seeks to construct a function F such that

$$(y, E, z) = F(X, R, K), \tag{1}$$

where R is a deterministic rulebase and K is an external or indexed knowledge source.

The architecture is guided by five objectives. First, it should separate learned inference from deterministic validation. Second, it should maintain provenance so that each important recommendation can be traced to an input feature, rule, model output, or retrieved evidence item. Third, it should support batch execution across multiple patient cases without mixing patient context. Fourth, it should produce a human-readable report instead of exposing raw model outputs. Finally, it should remain modular so that an individual agent or model can be replaced without redesigning the entire pipeline. These objectives translate into four design constraints:

- *Patient isolation:* no evidence, retrieved passage, or rule result from one case may be reused in another case unless it is part of the shared knowledge base.
- *Deterministic guardrails:* explicit thresholds and policies are executed by a rule component before generative synthesis.
- *Evidence grounding:* retrieved knowledge is stored with identifiers and source metadata so that supporting evidence can be inspected.
- *Human oversight:* the system provides recommendations for review and does not act as an autonomous diagnostic authority.

B. Proposed MedIntel Architecture Fig. 1 shows the proposed MedIntel architecture. Its components are described in the following subsections.

1) Input Ingestion and Preprocessing The first stage accepts patient data from CSV files, tabular datasets, or associated logs. A schema validator checks data types, missing fields, duplicate records, unit consistency, and identifier integrity. The normalized representation is converted into a patient-specific case object so that downstream agents consume a stable interface. Text fields can undergo lightweight cleaning and entity extraction, while numeric variables are preserved in their original semantic units before model-specific preprocessing.

2) Machine-Learning Analysis Agent The Machine Learning Analysis Agent is responsible for learned patterns that can be expressed as classification, risk scoring, or anomaly detection tasks. The specific algorithm is intentionally left configurable. Depending on the dataset, the implementation may compare logistic regression, random forest, gradient boosting, or another suitable baseline. The agent returns a prediction, class probabilities when available, feature importance or explanation metadata, and the model version used. This output is treated as evidence rather than as a final diagnosis.

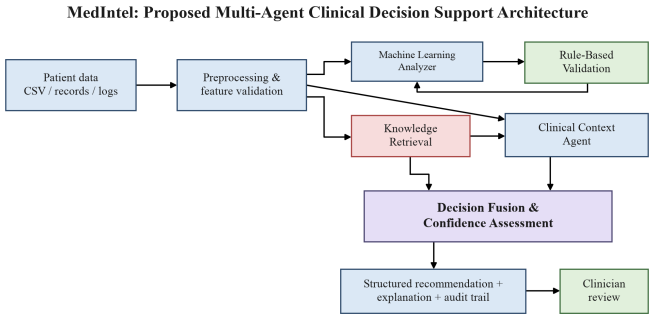


Fig. 1. Proposed MedIntel architecture combining learned inference, deterministic rule checks, retrieval-based evidence, and human review.

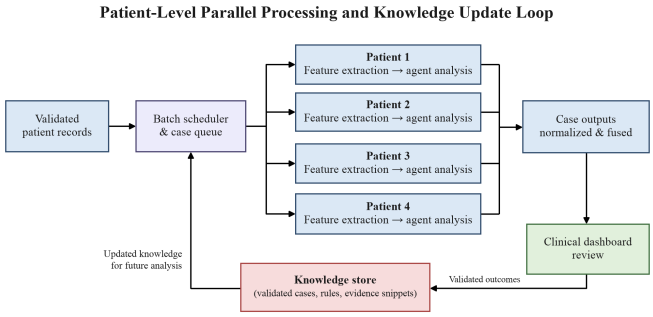


Fig. 2. Parallel case processing with isolated patient contexts and controlled knowledge-store updates.

3) Rule-Based Validation Agent The Rule-Based Validation Agent executes explicit clinical or project-defined conditions. For example, a rule can flag an input when a laboratory value exceeds a predefined threshold, when two findings occur together, or when a required field is missing. Rules are represented separately from the language model so that their behavior is deterministic and easy to audit. A rule may elevate, suppress, or flag a candidate recommendation, but it should not silently override a learned prediction without recording the reason.

4) Knowledge Retrieval Agent The Knowledge Retrieval Agent provides contextual evidence from an indexed medical knowledge collection. This design follows the broader evidence-grounding principle supported by RAG studies [6], [8], [9]. A practical implementation can store clinical articles, guideline excerpts, validated educational material, or institution-approved references in a vector index. A query derived from the current patient case is embedded and matched against the collection. Retrieved passages are returned with source identifiers and similarity scores, allowing the final response to state not only what was retrieved but also where it came from.

5) Clinical Context and Synthesis Agent The synthesis agent receives the outputs of the ML, rule, and retrieval components in a structured message. Its role is to reconcile the evidence, identify agreement or conflict, and generate a concise clinical-support report. The agent is instructed not to invent unsupported observations. Where the ML model and rule engine disagree, the conflict is reported explicitly. Where retrieval produces no sufficiently relevant evidence, the explanation should say so rather than filling the gap from unrestricted model memory.

6) Decision Fusion and Confidence Layer A conceptual confidence function can be defined as

$$C = \alpha M + \beta R + \gamma K - \lambda D, \tag{2}$$

where M is normalized machine-learning support, R is rule consistency, K is retrieval support, D represents disagreement among components, and the coefficients α, β, γ, and λ control the relative weight of each evidence source. This equation is not presented as a clinical standard; it is a design mechanism that makes the fusion logic explicit and testable. In the eventual implementation, the coefficients and thresholds should be learned or tuned only on a properly separated validation set.

7) Explanation and Audit Layer A dedicated explanation and audit stage is particularly important because modern reviews of clinical LLM systems identify calibration, validation, and transparency as recurring weaknesses [7], [12]. The final stage converts the machine-generated reasoning trace into a structured report containing: patient identifier, salient findings, model prediction, triggered rules, retrieved evidence references, conflicts, confidence indicators, and a short recommendation for clinician review. An audit record should retain the model version, timestamp, rule version, retrieval identifiers, and final status.

8) Multi-Patient Parallel Execution The system is designed to process several patient cases in parallel, as shown in Fig. 2. A batch scheduler places validated records into a queue and assigns each case an isolated context. The agents can then work concurrently where dependencies permit, while the fusion stage waits for all required evidence for a particular case. This structure makes the design suitable for hospital datasets containing many patients and also simplifies performance measurement because each case becomes an independent evaluation unit.

C. Implementation and Evaluation Protocol

Since MedIntel is currently an idea-level proposal, this section defines the experimental protocol that should be followed during implementation. The primary goal is not to maximize a single accuracy number, but to determine whether each architectural component contributes measurable value and whether the complete system remains reliable under controlled conditions.

1) Dataset Preparation The datasets available from the base papers [1], [2], along with sample datasets created by the authors, are the intended starting point. A suitable public clinical dataset should be selected according to the target task and documented before implementation. The data should be split into training, validation, and held-out test partitions without patient leakage. If the project uses multiple sources, identifiers and schemas must be normalized while preserving the origin of each record. Any de-identification or synthetic-data decision should be documented.

2) Baselines and Ablation The proposed system should be compared with progressively richer baselines. A minimal baseline can use the machine-learning model alone. A second condition can add the rule engine. A third can add knowledge retrieval, while the final condition uses the complete multi-agent fusion pipeline. Ablation experiments should remove one component at a time to identify whether the system depends excessively on any single agent.

3) Evaluation Metrics Task-specific metrics should include accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve when appropriate. Because a clinical-support system is more than a classifier, additional measurements should include calibration error, evidence retrieval precision at k, disagreement rate, false-alert rate for rules, end-to-end latency, and per-case processing cost. Human reviewers can also score explanation completeness, factual consistency, and usefulness on a defined rubric.

4) Reliability and Safety Checks The system should be evaluated on missing-data cases, contradictory evidence, borderline rule values, and deliberately ambiguous inputs [6], [12]. Each failure should be categorized as a data-validation error, model error, retrieval failure, rule conflict, synthesis error, or interface problem. This error taxonomy will make the final discussion

more informative than a single aggregate metric. Recent systematic evidence also supports emphasizing external validation and calibration rather than reporting benchmark accuracy alone.

5) Reproducibility Every experiment should record the dataset version, preprocessing configuration, model name and version, hyperparameters, rulebase version, retrieval index version, prompt template, random seeds where applicable, and evaluation script. The final implementation should be executable from a clean environment so that another researcher can reproduce the reported numbers.

IV. DISCUSSION

The main strength of MedIntel is the explicit separation of responsibilities. A machine-learning model can recognize statistical patterns without being asked to retrieve literature. A rule engine can enforce known thresholds without being asked to produce natural-language explanations. A retrieval component can ground a recommendation without assuming that retrieved text alone constitutes a diagnosis. The language model then acts as an interface between these signals, translating them into an explanation that a human can inspect.

This separation also addresses a practical weakness of highly agentic systems: when many LLM calls are chained together, errors can propagate between agents and it becomes difficult to identify where the final recommendation originated. In MedIntel, every major evidence source has a defined output contract. The fusion layer therefore has a clearer responsibility than an open-ended conversation among agents. This design is influenced by the modularity reported in MedAgents [2] and the evidence-grounding strategies explored by ClinicalRAG [3] and the 2025 adaptive CDSS study [1].

Another important consideration is computational cost. Multi-agent systems can require more inference calls, retrieval steps, and serialization overhead than a single predictive model. The proposed architecture therefore recommends parallelizing independent agents, reserving generative synthesis for the final stages, and using deterministic preprocessing whenever possible. This is consistent with the broader literature, which recognizes that improved grounding and collaboration can introduce additional latency and system complexity.

The current paper has a clear limitation: the MedIntel system has not yet been implemented, so no empirical claim about diagnostic accuracy, clinical safety, or superiority over existing methods can be made. The architecture is also intentionally generic. The most appropriate machine-learning model, retrieval strategy, rule syntax, confidence weights, and output schema will depend on the chosen dataset and clinical task.

Finally, the knowledge base should not be treated as an automatic memory that learns from every model output. Only validated and appropriately governed information should be re-indexed. This constraint is important because the knowledge-store update path can otherwise turn one incorrect recommendation into a persistent source of future errors. Recent work on self-learned clinical knowledge demonstrates the potential of iterative multi-agent learning, but it also makes the governance problem more visible [10].

V. CONCLUSION

MedIntel presents a practical architecture for multi-agent clinical decision support in which machine learning, deterministic rules, knowledge retrieval, and language-model synthesis have explicit and complementary roles. The design is intended to make patient-level reasoning more traceable while supporting parallel analysis of multiple records. Its central idea is not to replace established analytical methods with an LLM, but to orchestrate different sources of evidence and expose their contributions to human reviewers. The next step is empirical validation: implement the proposed pipeline, measure each component independently, conduct ablation and calibration studies, and evaluate the resulting system under realistic error and ambiguity conditions. Only after those experiments should performance claims be introduced into a final research manuscript.

Future work should first implement a reproducible prototype and evaluate it on a carefully selected benchmark. Subsequent work can investigate richer retrieval, medical ontologies, knowledge graphs, specialized language models, clinician-in-the-loop review, and longitudinal patient data. External validation across datasets or institutions would be particularly important because recent reviews continue to identify limited external validation and calibration as major barriers to clinical translation.

DECLARATIONS

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